Two-Block Circuit Formation: Stationary Theory versus Training

Result in one sentence. The fully trainable network can form a $2+2\to 4$ circuit, but only in a finite layerwise-decay window: weak first-block decay leaves a cheap block-1 shortcut, whereas strong first-block decay destroys the degree-2 parents; initialization determines whether training finds the compositional minimum inside that window.

Theory tested. For singleton norm $s = Q_1^{(0)}$, parent norm $q = Q_2$, and $Q_1^{(1)} = \sqrt{s^2+q^2}$, the reduced zero-temperature theory compares the optimized direct hybrid $H_D = D_1 + D_2$ with the compositional hybrid $H_C = D_1 + C$. At fixed representation, block 2 selects C precisely when

$$R \equiv Q_2/Q_1^{(1)} > 1/\sqrt{8}.$$

The representation is itself optimized, so the actual boundary follows from $F_C = F_D$. Inside H_C, the predicted compositional output share is

$$f_C = \frac{8\lambda_1 q^2}{\lambda_2 s^2 + 8\lambda_1 q^2}.$$

Increasing λ_1/λ_2 therefore suppresses the block-1 shortcut but also raises the cost of the block-1 parents, producing a bounded rather than monotone compositional region.

Experiment. We trained a bias-free two-residual-block ReLU network of width $N = 1024$ on the exact four-bit cube. Both degree-2 parents and the degree-4 target were trained simultaneously, with no frozen weights and $\gamma_2 = \gamma_4 = 2$. The tested plane was

$$\lambda_E = \lambda_2 = \lambda_p = \lambda_4 = \lambda_0, \qquad \lambda_1 = (\lambda_1/\lambda_2)\lambda_0,$$

using nine λ_0 values, eight ratios, three seeds, and two protocols. *Joint* trains both parent targets and the degree-4 target from the common random initialization, with every weight trainable. *Parent-warm* first trains E, V_1, W_{d2} on the parents read from h_1, then unfreezes everything and trains the identical final objective (216 runs per protocol; 432 total). Full-batch float64 SGD used $\eta_{E,V} = 10^{-2}$ and the muP readout scaling $\eta_W = 10^{-2}/N$. Each cell ran for at least 15k and at most 50k steps. Every empirical row below uses the protocol specified before training; no post-hoc selection between protocols is used.

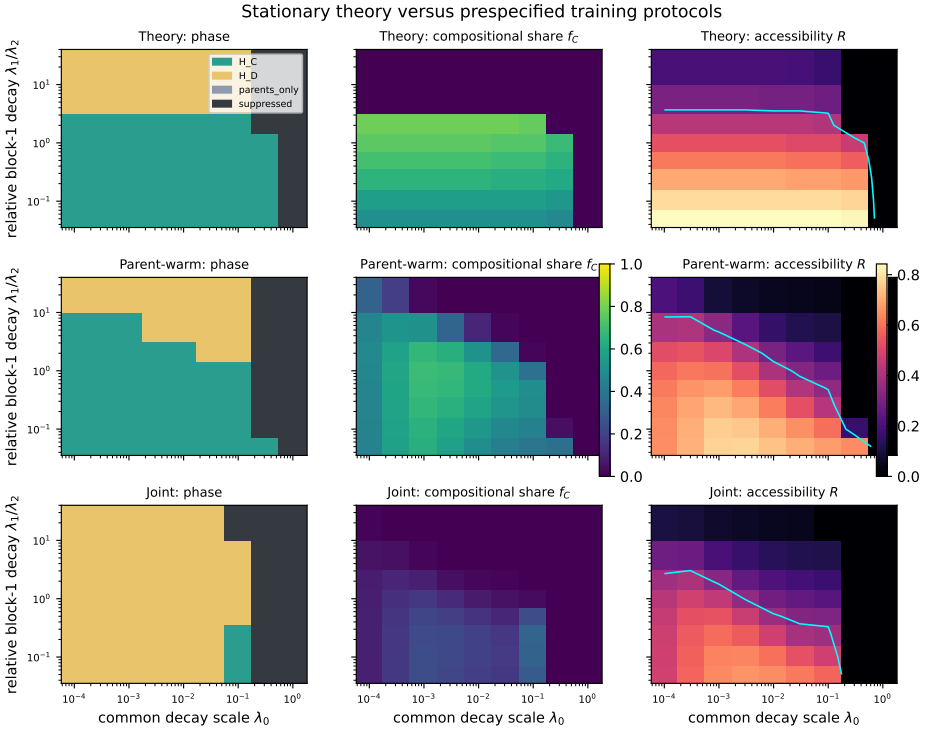


Figure 1: **Theory (top), parent-warm (middle), and joint training (bottom).** Columns show phase, total compositional target share, and final accessibility. Cyan marks $R = 1/\sqrt{8}$. Each empirical row averages its own three seeds; neither row is selected or replaced using the other protocol's final objective.

Quantitative comparison.

Diagnostic	Parent-warm	Joint
Theory/protocol phase agreement over the 8×9 grid	87.5%	37.5%
Measured-R gate agreement in active cells	91.2%	49.1%
Mean absolute error in total compositional share f_C	0.152	0.337
Grid cells with majority phase H_C	44/72	3/72
Individual runs classified H_C	134/216	9/216

The two protocols select different phases in 59.3% of matched seed/cell comparisons (128/216).

What the phase diagram establishes. First, the desired circuit is feasible without freezing: the prespecified parent-warm protocol occupies a substantial band at moderate λ_0 and moderate λ_1/λ_2, and reproduces the theoretical phase topology much better than joint training. Second, the two failure modes predicted by the reduced theory are distinct. At small ratio, parents remain large but the degree-4 readout uses the inexpensive D_1 shortcut. At large ratio, D_1 is suppressed but Q_2 collapses, R falls below $1/\sqrt{8}$, and block 2 uses D_2 or the target is suppressed. Third, matched joint and parent-warm runs—same seed, same decay cell, and identical final objective—selected different phases in 128 of 216 cases. This hysteresis is finite-time/nonconvex basin selection, not a shift of the local gate boundary. The largest parent-warm mean compositional share was 0.681 at $\lambda_0 = 10^{-3}$ and $\lambda_1/\lambda_2 = 1$, with $R = 0.595$ and target amplitude 0.995; the theory selects H_C in the same cell and predicts $f_C = 0.749$.

How to favor $2 + 2 \to 4$. Use nonzero decay on the degree-4 readout so hidden/readout rescaling cannot evade the norm comparison. Keep λ_E, λ_p, and especially the parent-producing part of λ_1 small enough that the *final* measured R stays above $1/\sqrt{8}$; then make block 2 cheaper than the block-1 degree-4 shortcut by lowering λ_2 relative to the target-specific block-1 decay. Whole-layer λ_1 is an imperfect control because it taxes wanted parents and the unwanted shortcut together. If the feasible window is narrow, separate those directions with a decayed block-1 degree-4 adapter or soft subspace penalty.

Interpretation and limits. The stationary theory captures the topology of the control problem—a compositional window between shortcut domination and parent collapse—but quantitative differences are expected. The finite network contains mixed Walsh features, non-orthogonal singleton leakage, and multifunctional neurons omitted by the balanced rank-one ansatz. Moreover, many runs hit the 50k-step cap (333 of 432), while protocol disagreement directly demonstrates metastability. Thus the parent-warm map is a history-conditioned accessibility map, not proof of equilibrium, while the joint map directly displays the optimization barrier from random initialization. The most robust local diagnostic remains the measured inequality $Q_2/Q_1^{(1)} > 1/\sqrt{8}$, together with the signed target contributions from h_1 and the second residual branch.

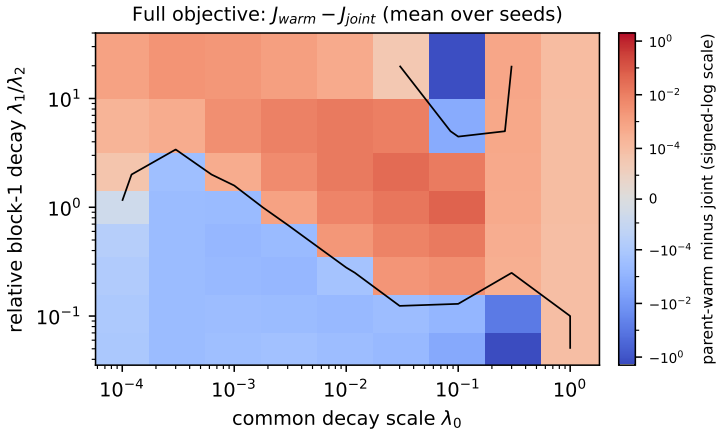


Figure 2: **Full regularized objective difference between the two prespecified protocols.** Each cell shows the mean matched-seed difference $J_{\text{warm}} - J_{\text{joint}}$ using a signed-log color scale: blue favors parent-warm, red favors joint, and the black contour marks equality. This is an objective diagnostic only; it is not used to select or replace either empirical phase-map row in Fig. 1.